

1. Explain the role of Query Editor in Power BI and how it helps in preparing data before creating reports.

Query Editor, also called Power Query Editor, is a tool in Power BI used to connect, clean, and transform data before it is used to create reports and visualizations.

It helps to:

- Remove blank and duplicate rows.
- Change data types such as text, number, and date.
- Filter unwanted data.
- Rename columns.
- Split or merge columns.
- Replace incorrect or missing values.
- Merge and append tables.
- Create new columns.
- Sort and organize data.

After completing the transformations, the prepared data is loaded into the Power BI data model and can be used to create reports.

In simple words:

Query Editor → Clean & Transform Data → Load Data → Create Reports

2. What is data transformation in Power BI, and why is it critical for accurate analysis and reporting?

Data transformation is the process of cleaning, modifying, and converting raw data into a suitable format for analysis in Power BI.

For example, raw data may contain blank rows, duplicate records, incorrect data types, inconsistent names, or unnecessary columns. Data transformation helps correct these problems.

Common transformations include:

- Removing duplicates and blank rows.

- Changing data types.
- Filtering rows.
- Splitting and merging columns.
- Replacing values.
- Removing unnecessary columns.
- Combining multiple tables.
- Creating calculated or conditional columns.

Why is it critical?

Data transformation is important because poor-quality data can produce incorrect reports and misleading business insights.

For example, if a sales column is stored as text instead of numbers, Power BI may not calculate total sales correctly. Similarly, duplicate records can result in inflated sales figures.

Therefore, proper data transformation ensures that the data is clean, consistent, accurate, and ready for reliable analysis and reporting.